

Assignment 2: QSAR Data Curation

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1. Introduction: Objective and Target Background

The objective of this assignment was to demonstrate a complete and reproducible workflow for QSAR data collection and curation using publicly available bioactivity data. Specifically, the task focused on retrieving, preprocessing, and organizing high-quality bioactivity data suitable for downstream QSAR modeling.

The selected biological target for this study is RAD51 (DNA repair protein RAD51 homolog 1), a key protein involved in homologous recombination and DNA double-strand break repair. RAD51 plays a crucial role in maintaining genomic stability and has been widely investigated as a therapeutic target in cancer, where its dysregulation contributes to tumor progression and resistance to chemotherapy and radiotherapy. Due to its biomedical relevance and availability of experimental data, RAD51 represents a suitable target for QSAR analysis.

2. Target and Data Selection

- Selected target name: DNA repair protein RAD51 homolog 1
- ChEMBL Target ID: ChEMBL2034807 (*Homo sapiens*)

Data Evaluation and Justification

Bioactivity data were retrieved from the ChEMBL database, a curated and widely used repository of experimentally validated bioactivities. IC₅₀ values were selected as the activity endpoint because they are quantitative, standardized, and commonly used in QSAR modeling.

- **Number of available bioactivity records (IC₅₀):** Retrieved directly from ChEMBL for the selected RAD51 target. The unit was in Nanomolar (nM)

The choice of IC₅₀-based bioactivity data ensures consistency and suitability for regression or classification-based QSAR modeling.

3. Data Curation Workflow: Retrieval and Preprocessing

The data curation process followed best practices for QSAR dataset preparation:

1. **Environment setup:** Google Colab was connected to Google Drive to enable persistent storage of datasets across sessions.
2. **Target identification:** RAD51 was searched in ChEMBL, and the most relevant human target entry was selected.
3. **Bioactivity retrieval:** All bioactivity records associated with the selected target were retrieved and filtered to include only IC₅₀ measurements.
4. **Data cleaning:** Removed records with missing or invalid IC₅₀ values and excluded compounds lacking valid canonical SMILES strings
5. **Bioactivity classification:** Compounds were categorized based on IC₅₀ values to active, intermediate and inactive
6. **Dataset construction:** A final curated dataset containing molecule ChEMBL ID, canonical SMILES, IC₅₀ values, and bioactivity class was generated and saved as a CSV file.

4. Workflow Overview

Step-by-Step Summary

1. Mount Google Drive in Google Colab
2. Install and import required Python libraries
3. Search and select the RAD51 target in ChEMBL
4. Retrieve IC₅₀ bioactivity data
5. Filter and clean raw bioactivity records
6. Assign bioactivity classes
7. Generate and save the preprocessed QSAR-ready dataset

Conclusion

This assignment successfully demonstrates a complete QSAR data curation pipeline, transforming raw ChEMBL bioactivity records into a clean, structured, and modeling-ready dataset. The curated RAD51 bioactivity dataset provides a reliable foundation for subsequent descriptor calculation and QSAR model development. Please find my GitHub link at:

https://github.com/RajiAbdulrahman/AI_and_Drug_Discovery_Course_2026